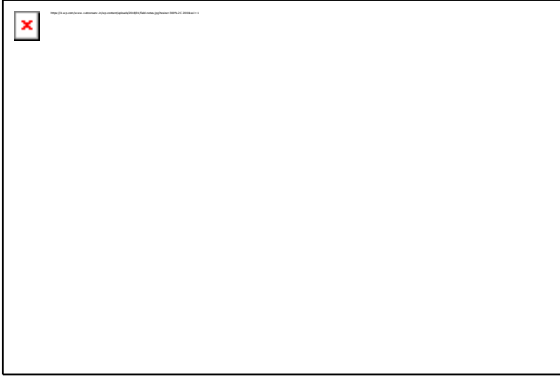


PM Guide for field Veterinarians

When we write a post-mortem report on a field we arrive at the tentative diagnosis. However, we can not recall the standard set of lesions present in that particular condition. Therefore, this table will be a great troubleshooter.

It is not necessary that all the lesions mentioned in the table must be present in the case you attaining, so the modification is obviously required during such conditions.



Index

1. Lightning stroke
2. Electric shock/current
3. Gun shot
4. Fell down due to slippery ground
5. Drowning
6. Poisoning
7. Road accident
8. Hemorrhagic septicaemia
9. Black quarter
10. Snake bite- neurotoxic type (cobra and krait)
11. Snake bite- hemolytic type (viper)
12. Heat stroke/ Sun stroke
13. Acute copper poisoning
14. Subacute/ chronic copper poisoning
15. Acute aflatoxicosis in calves
16. Subacute aflatoxicosis in calves
17. HCN/Cynide/ young jowar shoots poisoning
18. Gossypol/ cottonseed cake poisoning
19. Lead poisoning
20. Salt poisoning
21. Urea poisoning

22. Collibacillosis
23. Rinder pest
24. Foot and mouth disease
25. Traumatic pericarditis
26. Johne's disease/Paratuberculosis
27. Rabies
28. Naval ill in calf
29. TRP
30. Theileriasis
31. Babesiosis
32. Surra
33. Anaplasmosis
34. Coccidiosis (calf)
35. Ketosis
36. Grass tetany
37. Downer cow syndrome (milk fever)

Sr	History	Set of lesions
1	Lightening stroke	Very rapid rigor mortis
		Skin- tree shaped branching often reddish or bluish in colour is seen
		Singed hairs with severe and deep burns present may reach upto the bones
		Blood- blackish and unclotted
		Lungs- congested and oedematic
		Congestion and hemorrhages on majority of visceral organs
2	Electric shock/current	Skin- mark of burn where animal contacted with electric wire, surrounding hairs burned/scorched, peripheral lacerated radiating mark
		Subcutaneous tissue-deep burns at affected area
		Nasal cavity- reddish coloured fluid oozing
		Oral cavity- half chewed food
		Trachea- petechial hemorrhages

Sr	History	Set of lesions
		Lungs-congested and oedematous
		Heart-petechial hemorrhages on surface, dark (black) coloured unclotted blood in heart chambers, dilated chambers
		Liver-congested
		Kidneys-congested
		Lymph nodes-petechial hemorrhages
3	Gun shot	Entry wound-On the skin characterised by lacerated 2-3 cm opening and its borders showing clotted blood.
		Exit wound- present -On opposite side if fired from short distance Exit wound- absent -if fired from long distance and a metallic object(bullet) may be present in body of animal in vicinity of entry wound
		Subcutaneous tissue- around the wound congested
		Oral cavity- reddish coloured fluid oozing (may be)
		Nasal cavity- reddish coloured fluid oozing (may be)
		Body cavities- peritoneal or thoracic cavity show clotted blood depending on affected part of body
		Visceral organs- rupture or damage depending on site of bullet injury
4	Fell down due to slippery ground	Skin abrasions/contusions
		Reddish coloured fluid may be oozing from nasal openings
		Subcutaneous tissue below affected skin shows severe congestion
		Fracture of bones may be present depending on part of body hit ground
		Rupture of organs such as liver/spleen with clotted blood on the margins
		Massive reddish fluid/clotted blood in the body cavities
		Heart- empty ventricles
5	Drowning	Presence of multiple folds on skin (corrugated appearance)

Sr	History	Set of lesions
		Firm and leathery appearance of muscles
		Abrasions and wounds on body surfaces
		Distended abdomen
		Oral cavity- frothy exudate
		Nasal cavity- frothy exudate
		Lungs- highly distended, on section a clear fluid (water) oozing from incision
		Heart- semiclotting blood in left ventricle and veins contains partially clotted blood
		Digestive tract- rumen and reticulum filled with excess of fluid (water)
6	Poisoning	Distended abdomen
		Trachea- presence of fluid/frothy material
		Lungs- congested
		Heart- clotted blood
		Liver- hepatomegaly
		Digestive tract- congestion and hemorrhages on mucous membrane of rumen and intestines. Reddish coloured ingesta in intestinal lumen
		Lesions suggestive of death occurred due to severe gastritis and intestinal hemorrhages may be caused by poisoning
7	Road accident	Skin with abrasions and contusions. Skeletal muscles hemorrhagic and contains blood clots within it
		Fracture/dislocation of limbs/horn/ribs
		Nasal and oral cavity- blood containing fluid oozing
		Bloody fluid/clots in thoracic and abdominal cavity
		Heart- empty with dilated right side
		Rupture of liver/spleen with blood clots

Sr	History	Set of lesions
8	Hemorrhagic septicaemia	Edematous swelling on neck region
		Trachea- lumen filled with frothy exudate
		Lung- congestion, consolidation and fluid in bronchi and bronchioles
		Heart-petechial hemorrhages on surface and chambers with clotted blood
		Liver- petechial hemorrhages
		Spleen- petechial hemorrhages
		Spleen- petechial hemorrhages
		Digestive system- congested mucosa
		Kidneys- petechial hemorrhages
9	Black quarter	Crepitating swelling of thigh and shoulder muscle. On incision muscles appear as necrosed and black in colour. Fluid containing gas bubble present in affected muscle giving rancid odour.
		Liver- swollen
		Spleen- swollen
		Lymph nodes- swollen
		Heart- presence of clotted blood
10	Snake bite- neurotoxic type (cobra and krait)	Bite marks may present on body often on hairless part of skin (muzzle)
		Lungs- oedematic and congested
		Heart- right chambers of heart dialated
11	Snake bite- hemolytic type (viper)	Bite marks may present on body often on hairless part of skin (muzzle)
		Unclotted blood
		Body cavities shows unclotted reddish fluid
		Absence of bloating (balloning after death)
12	Heat stroke/ Sun stroke	Skin- reddened

Sr	History	Set of lesions
		Poor blood coagulation
		Heart- dilatation of right side
		Congestion, hemorrhages and oedema of visceral organs (liver, kidneys, lungs, brain)
13	Acute copper poisoning	Digestive tract-presence of blue-green ingesta and severe gastroenteritis with ulceration and erosions in abomasal mucosa
		Liver and kidneys degenerative changes
14	Subacute/ chronic copper poisoning	Tissue discoloration due to icterus (yellowish) and methemoglobin formation (chocolate brown)
		Liver- swollen, friable and icteric. Gall bladder distended with excess of thick, yellow bile
		Spleen- enlarged, friable with pulpy dark brown black parenchyma
		Kidneys- swollen and gunmetal-colored
		Urinary bladder- port wine coloured urine
15	Acute aflatoxicosis in calves	Liver - hepatitis
		Hemorrhages on internal organs with icteric appearance
16	Subacute aflatoxicosis in calves	Liver- enlarged with firm appearance
		Gall bladder- edematous
17	HCN/Cyanide/ young jowar shoots poisoning	Cynotic mucous membranes
		Heart- right chambers contains bright cherry red coloured blood
		Clotting of blood is either slow or absent
		Digestive tract- rumen is highly distended and on opening a smell resembling bitter almond is evident
18	Gossypol/ cottonseed cake poisoning	Heart- enlarged, pale, mottled with myocardial streakings. Dilated ventricles with edematous valves
		Skeletal muscles-pale

Sr	History	Set of lesions
		Lungs- edematic with frothy material in tracheal lumen and interlobular septa.
		Liver- Icteric, enlarged, congested, mottled and golden in colour. Characteristic nutmeg appearance.
		Kidneys, spleen- congested with petechial hemorrhages
19	Lead poisoning	Digestive tract- presence of oil paint or residue of battery in lumen. Severe gastroenteritis
		Brain- congestion and edema of cerebral cortex
		In my opinion present animal has died due to gastritis and damage to nervous system suggestive of lead toxicity.
20	Salt poisoning	Digestive tract- mild to severe gastric congestion with hemorrhages and ulcerations
		Parched (very dry) appearance of gastic content
21	Urea poisoning	Severe bloat
		Presence of large amount ruminal fluid on ground near the nasal opening
		Presence of white foamy exudate in airways
		Generalised congestion of carcass
		Heart- Hydropericardium, petechia on epi and endocardium
		A ruminal content gives ammonical odour after opening of rumen
		Lunges- pulmonary edema
		Ruminal fluid gives alkaline reaction when tested with the pH paper (7.5-8)
22	Collibacillosis	Carcass highly emaciated and anaemic
		Parched and dry tongue
		Heart- partially clotted blood in left ventricle
		Liver- swollen with thick bile in gall bladder
		Digestive tract- congested mucosae of abomasum and intestines with

Sr	History	Set of lesions
		mucooid ingesta
		Kindeys-congested
23	Rinder pest	Emaciated carcass, with hemorrhages and pustules on skin.
		Conjuctivitis
		Oral cavity- greyish white nodules and mucopurulent discharge
		Nasal cavity- mucopurulent discharge
		Lungs- congested and emphysematous
		Heart- clotted blood
		Liver- hepatomegaly with petechial hemorrhages on surface
		Digestive tract- congested and hemorrhagic mucous membranes of abomasm and intestine. Necrosed Payer's patches. Charecteristic zebra marking on colon and rectal mucosa.
		Mesenteric lymph nodes- swollen, congested and hemorrhagic
		Kidneys- renomegaly
24	Foot and mouth disease	Presence of vesicular lesions on the lips, dorsal side of tongue, skin of coronary band and interdigital space, teats
		Heart- tigroid condition in young animals
		Digestive tract- congested and edematic mucous membrane of abomasum
25	Traumatic pericarditis	Anaemia-pale mucous membranes
		Oronasal cavity- reddish coloured fluid oozing
		Heart- enlarged heart. Blood stained thickned fibrous appearance of pericardium. Pericardial sac distended containing fibrinopurulent material.
		Adhesions between the heart, reticulum and diaphragm with presence of mettalic object/nail/wire penetrating from reticum to heart through diaphragm
		Digestive tract- Reticulitis with marked adhesions with surrounding

Sr	History	Set of lesions
		tissues. Presence of metallic wire/ nail stuck in wall of reticulum
26	Johnes disease/Paratuberculosis	Severe emaciated and anaemic carcass
		Heart- clotted blood
		Digestive tract- pale, oedematous and thickened mucosa of ileum and caecum. Corrugated appearance of mucous membrane with severe congestion
		Lymph nodes- mesenteric lymph nodes swollen
		In my opinion lesions are suggestive of death due to Johnes disease (Paratuberculosis)
27	Rabies	Multiple abrasions, contusions and injuries on body particularly on head region
		foreign material in digestive tract
		Reddened conjunctiva
		Brain- congestion
28	Naval ill in calf	Emaciated and anaemic carcass
		Umbilical regions showing sever swelling and on incision lesions shows purulent material mixed with bloody exudate
		Lungs- congested consodited and pneumonic
		Peritoneal cavity- reddish fluid
		Liver- swollen with suppurative lesions on surface
		Gall bladder distended with accumulated thick bile fluid
		Kidneys- congested and swollen
		Lymph nodes- mesenteric lymph nodes highly congested
29	TRP	Severe emaciation
		Subcutaneous fat-gelatinization
		Pericardium- presence of fluid in pericardial sac forming fibrinous

Sr	History	Set of lesions
		strands
		Coronal fat- jelly like
		Peritoneal cavity- Presence of serous fluid
		Spleen- multiple abscesses, spleen shows adherence with surrounding tissue/diaphragm, A wire/nail/metallic object stuck in the spleen
		Reticulum-presence of a metallic object/nail/wire penetrating through its wall
30	Theileriasis	Severe emaciation
		Anaemia-pale mucous membranes
		Subcutaneous and intramuscular edema
		Subcutaneous fat-gelatinization
		Petechial hemorrhages on tongue
		Pleural cavity/lungs- Presence of excessive fluid in pleural cavity. Lungs congested, emphysematous and edematous
		Heart- Hydropericardium, petechial hemorrhages on endocardium
		Liver-hepatomegaly, congested with petechial hemorrhages
		Gall bladder-petechial hemorrhages on surface
		Spleen- Splenomegaly and congestion
		Abomasum- punched out ulcers with petechial hemorrhages on mucosa
		Kidneys-congested
		Brain-focal hemorrhages
		Prescapular lymph node-highly enlarged and congested
31	Babesiosis	Severe emaciation, anaemic and icteric appearance
		Subcutaneous tissue shows icteric condition and adipose tissue gelatinized
		Heart-clotted blood in chambers

Sr	History	Set of lesions
		Peritoneal cavity- Presence of yellowish fluid
		Liver-enlarged with yellowish brown colour
		Gall bladder- distended with bile
		Spleen- severe splenomegaly with soft and dark appearance
		Kidneys-swollen and ecteric
		Urinary bladder-presence of red coloured urine
32	Surra	Emaciated carcass, severe anaemia, edematic condition, muscular atrophy
		Subcutaneous edema
		Lungs- congested
		Peritoneal cavity- presence of serous fluid
		Liver- hepatomegaly with petechial hemorrhages on surface
		Kidneys-swollen with petechial hemorrhages on surface
		Brain-congested
33	Anaplasmosis	Emaciated carcass, severe anaemia, edematic carcass, muscular atrophy
		Icteric mucous membranes
		Subcutaneous tissue- icteric and gelatinization of adipose tissue
		Lungs- emphysematous
		Heart- dialated with clotted blood in chambers
		Peritoneal cavity-presence of large amount of yellowish fluid
		Liver- hepatomegaly with distended gall bladder containing dark and viscid bile
		Splenomegaly
		Kidney- yellowish and straw coloured fluid in urinary bladder
		Lymph nodes-edematic

Sr	History	Set of lesions
34	Coccidiosis (calf)	Emaciated carcass, sunken eyes, posterior part of body soiled with faeces mixed with blood
		Heart- chambers containing clotted blood
		Digestive tract- mucous membrane of small and large intestines highly congested, hemorrhagic and lumen filled with material containing blood
		Mesenteric lymph nodes- congested and edematic
35	Ketosis	Emaciated carcass with anaemia
		Subcutaneous tissue- jelly like appearance of adipose tissue
		Heart- clotted blood
		Peritoneal cavity- contains watery fluid
		Liver- hepatomegaly with bright yellow colour, cut surface shows greasy appearance (may float on water due to severe fatty changes). Gall bladder distended with bile
		Urinary bladder- urine contains characteristic smell of ketone bodies (sweet smell)
36	Grass tetany	Contracted and hardened muscles
		Subcutaneous tissue hemorrhagic
		Oral cavity-frothy fluid
		Lungs- emphysematous and pleura- hemorrhagic
		Heart- hemorrhages on pericardium and endocardium. Pericardial sac showing serosanguinous fluid. Raised, calcified areas on intima (inner walls) of large vessels.
		Digestive tract- segmentally contracted abomasum and small intestine. Hemorrhagic mucous membrane of rumen and small intestine
37	Downer cow syndrome (milk fever)	Skin abrasions, thigh muscle showing hemorrhages and necrotic changes
		Heart- empty and dilated ventricles, degenerative changes in myocardium
		Peritoneal cavity- excess serous fluid

Sr	History	Set of lesions
		Liver- hepatomegaly and fatty changes
		Kidneys- swollen (renomegaly)